

Scaling Book Exercises: Section 9

Lucas Yan

Chapter: How to Profile TPU Programs

Source scan: ../handwritten/section_09_q1_handwritten.pdf

Reference: <https://jax-ml.github.io/scaling-book/profiling/>

Question 1: Reading the suspicious profile

The profile contains a reduction, two large fusions containing matrix multiplications, and a final AllReduce. The underlying computation is

$$\text{out} = \text{einsum}(\text{wf}, \text{bf} \rightarrow \text{bw}, W_2, \text{einsum}(\text{fw}, \text{bw} \rightarrow \text{bf}, W_1, x)).$$

Equivalently, this is the two-layer linear map

$$h = xW_1^T, \quad \text{out} = hW_2^T.$$

First fusion

The HLO shows the local computation

$$\text{bf16}[8192] \times \text{bf16}[4096, 8192] \longrightarrow \text{bf16}[4096],$$

contracting over the width dimension 8192. The final AllReduce has replica groups of size eight, revealing eight-way model parallelism. Therefore the feed-forward dimension is sharded eight ways:

$$F_{\text{local}} = 4096, \quad F = 8(4096) = 32768.$$

Restoring the batch dimension, the true first matmul is

$$\boxed{x[8, 8192] W_1^T[8192, 32768] \longrightarrow h[8, 32768]}.$$

Every model-parallel device computes 4096 columns of h .

Second fusion and AllReduce

For one batch element, each device multiplies its local hidden vector by the corresponding slice of W_2 :

$$h_y[4096] W_{2,y}^T[4096, 8192] \longrightarrow p_y[8192].$$

Restoring the batch dimension, the same local computation is

$$h_y[8, 4096] W_{2,y}^T[4096, 8192] \longrightarrow p_y[8, 8192].$$

These are partial sums over disjoint pieces of the contracted feed-forward dimension. The AllReduce adds them:

$$\text{out} = \sum_{y=0}^7 p_y \in \mathbb{R}^{8 \times 8192}.$$

Thus the AllReduce is required because the sharded axis is a contracting axis in the second matmul; it is not an AllGather.

Profile operation	Local meaning	Global meaning
Reduction	preprocessing feeding x	not the model-parallel combine
First fusion	$8192 \rightarrow 4096$ matmul	$8192 \rightarrow 32768$ up-projection
Second fusion	$4096 \rightarrow 8192$ matmul	partial down-projection
AllReduce	sum eight partial outputs	final $[8, 8192]$ result

Conclusion. The suspicious-looking small 4096 dimension is only the per-device view. The true feed-forward width is 32768, split over eight devices, and the output is reconstructed by summing the eight partial down-projections.

Question 2 status. No benchmark result is reported here. The exercise requires a real eight-core TPU topology for meaningful timings; the available CPU fallback can demonstrate sharding semantics but cannot reproduce TPU performance.